NLP Project: Analysis and Implementation of SelfCheckGPT

Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models

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1 Introduction

LLMs have demonstrated remarkable ability to generate fluent and convincing texts on various topics and for several contexts. However, they are also capable of outputting misleading or even false information; this phenomenon is commonly referred to as hallucination.

In order to detect those hallucinations, most methods use token-level probability distributions to identify uncertain outputs. These "Grey-Box" methods require internal access to the model, which is often not guaranteed when evaluating top-end models like ChatGPT. One can also use internal databases, but these methods are restricted to the content of the said databases.

In this report, we analyze SelfCheckGPT (Manakul et al., 2023), a zero-resource, black-box hallucination detection method that requires neither internal model access nor external knowledge bases. The method leverages the consistency of multiple stochastically sampled responses to determine whether a given statement is factual or hallucinated. We analyze its contributions, evaluate its experimental setup critically, and present our own small-scale experiments exploring one of its variants.

2 Paper Analysis and Explanation

The central contribution of SelfCheckGPT is a simple but powerful observation: a well-trained LLM will produce consistent responses about facts it has learned, while hallucinated content will vary unpredictably across samples. By drawing multiple stochastic responses to the same prompt and measuring their mutual consistency, one can assign a hallucination score to each sentence in the original response, without any external resource. This makes SelfCheckGPT the first zero-resource hallucination detection method applicable to black-box systems.

2.1 Consistency evaluations

SelfCheckGPT proposes five variants for measuring consistency between a generated sentence and a set of stochastically sampled responses. Each variant offers a different trade-off between computational cost and detection performance.

N-gram The simplest variant builds a lightweight language model directly from the N sampled responses. Given this n-gram model, each token in the evaluated sentence is assigned a probability based on how frequently it (and its preceding context) appeared across the samples. A sentence containing tokens that rarely appear in any sample is likely hallucinated. The paper explores two scoring strategies, either averaging the negative log-probabilities of all tokens in the sentence, or taking the maximum. It finds that the maximum strategy is surprisingly effective, suggesting that a single anomalous token is often a reliable signal of hallucination. As n increases beyond 1, performance degrades, likely because longer n-grams become too sparse to estimate reliably from only 20 samples.

BERTScore This variant measures semantic similarity using contextual embeddings. For each evaluated sentence, the method finds the most semantically similar sentence in each sample using BERTScore and averages these similarity scores across all N samples. The hallucination score is then defined as one minus this average similarity. The intuition is that a factual sentence should find a close match in most samples, while a hallucinated sentence will be semantically distant from what the model generates in other runs.

Question Answering The QA variant takes a more indirect approach to consistency measurement. For each evaluated sentence, multiple-choice questions are automatically generated using the MQAG framework. These questions are then posed to an answering model conditioned on each of the N samples as context. If the samples are consistent with the evaluated sentence, the answering model should select the same answer as when conditioned on the original response. A mismatch signals potential hallucination.

NLI This variant relies on a model specifically trained to determine whether one piece of text contradicts another. The idea is straightforward: each sampled response is treated as a reference, and the evaluated sentence is checked against it. If the sample contradicts the sentence, this is a signal that the sentence may be hallucinated. Concretely, a DeBERTa-v3-large model is used to produce a contradiction score for each sample, and these scores are averaged across all N samples to give a final hallucination score. The neutral outcome is ignored, so the score reflects only how strongly the samples support or contradict the sentence. This variant performs very close to the Prompt variant in terms of detection accuracy, while being much cheaper to run, it does not require any API calls to a large LLM.

Prompting The most powerful variant queries an LLM directly to assess consistency. For each evaluated sentence and each sample, the LLM is prompted with the sample as context and asked whether the sentence is supported by it, answering Yes or No. The hallucination score is the proportion of samples for which the answer is No. This approach leverages the LLM's own language understanding and world knowledge to perform nuanced consistency assessments that go beyond what similarity metrics or NLI models can capture. In practice, GPT-3 outputs a valid Yes/No answer 98% of the time, with remaining outputs assigned a neutral score of 0.5. The method achieves the best performance across all evaluation settings, but comes at a steep cost: evaluating all sentences against 20 samples requires a large number of calls, making it expensive at scale. When trying with less expensive models such as LLaMA, performance drops significantly.

2.2 Experiments

The authors evaluate SelfCheckGPT on a dataset they construct themselves, since no standard hallucination detection benchmark existed at the time. They use GPT-3 to generate 238 Wikipedia-style passages about individuals from the WikiBio dataset, and manually annotate each sentence as either accurate, minor inaccurate, or major inaccurate, resulting in 1908 annotated sentences. Two evaluation tasks are considered: sentence-level hallucination detection, which measures how well each variant identifies non-factual sentences (using AUC-PR), and passage-level factuality ranking, which measures how well the passage-level scores correlate with human judgements (using Pearson and Spearman correlation). All five SelfCheckGPT variants are compared against grey-box baselines using GPT-3's own token probabilities, and black-box baselines using proxy LLMs such as LLaMA. Additional ablation studies investigate the effect of the number of samples, the choice of LLM for the Prompt variant, and the potential benefit of substituting self-samples with external knowledge from WikiBio. Across all settings, SelfCheckGPT consistently outperforms the baselines, with the Prompt and NLI variants achieving the strongest results.

3 Contribution

3.1 Implementation

We re-implement the SelfCheck-n-gram detector from scratch in pure Python, with no external NLP dependency (only `scikit-learn` and `scipy`, for the metrics). Texts are lower-cased and split by a simple regular expression keeping alphanumeric runs as words and isolating punctuation. For a given order n, we fit an n-gram model on the N stochastic samples concatenated with the response, padding each text with start/end symbols and counting, for every context c, the following token. Probabilities use add-k (Laplace) smoothing,

$$\hat{p}_n(w_j \mid c) = \frac{\mathrm{count}(c, w_j) + k}{\sum_w \mathrm{count}(c, w) + k\,|V|}, \tag{1}$$

with $k = 1$ and a uniform back-off $1/|V|$ for unseen contexts, so no token gets zero probability.

The hallucination score of a sentence $s_i = (w_1, \ldots, w_{J_i})$ is its surprise under the model; we implement both aggregators of the paper, the mean and the maximum per-token negative log-likelihood:

$$\mathrm{score}_{\mathrm{avg}}(s_i) = -\frac{1}{J_i}\sum_{j=1}^{J_i} \log \hat{p}_n(w_j \mid c_j), \qquad \mathrm{score}_{\mathrm{max}}(s_i) = \max_j \big(-\log \hat{p}_n(w_j \mid c_j)\big). \tag{2}$$

The passage score is the mean of its sentence scores. We follow the paper's protocol: sentence annotations are binarised into *NonFact* (label $\neq$ `accurate`) and *Factual*, scored by AUC-PR; passage annotations are mapped to $\{0, 0.5, 1\}$, averaged per passage, and compared to the passage score by Pearson and Spearman correlations.

3.2 Experimental results

Reproduction. On the official `wiki_bio_gpt3_hallucination` benchmark (238 passages, 1908 sentences), our cheapest variant — unigram with the `max` aggregator and $N = 20$ — closely matches the paper's published *Unigram (max)* numbers (Table 2 of the paper) on all four metrics: AUC-PR NonFact 86.9 vs. 85.6, Factual 58.9 vs. 58.5, Pearson 65.0 vs. 64.7, and Spearman 65.9 vs. 64.9. This confirms that our from-scratch implementation faithfully reproduces the n-gram variant. We then run three ablations not in the paper.

Aggregator and n-gram order. The `max` aggregator dominates `avg` on every metric (Fig. 1, left): a single highly surprising token signals hallucination better than the sentence-averaged surprise. Bigrams (Fig. 1, right) marginally help passage correlation but hurt Factual AUC-PR, as bigram contexts are too sparse at $N = 20$; unigram is the better default.

Number of samples. Since each sample is one extra LLM call, N is the main cost knob. Sweeping $N \in \{1, 2, 4, 8, 16, 20\}$ (Fig. 2, left) shows quality rising sharply up to $N \approx 8$ then plateauing: one can halve the cost from $N = 20$ to $N = 8$ at negligible loss, a practical point the paper omits.

Transferability to GPT-2. As an experiment not in the paper, we test whether the detector is generator-agnostic by replacing GPT-3 with `gpt2` (124M). We score GPT-2 biographies of two real figures and one fabricated name (*Zorgblax Quintarius*). All three

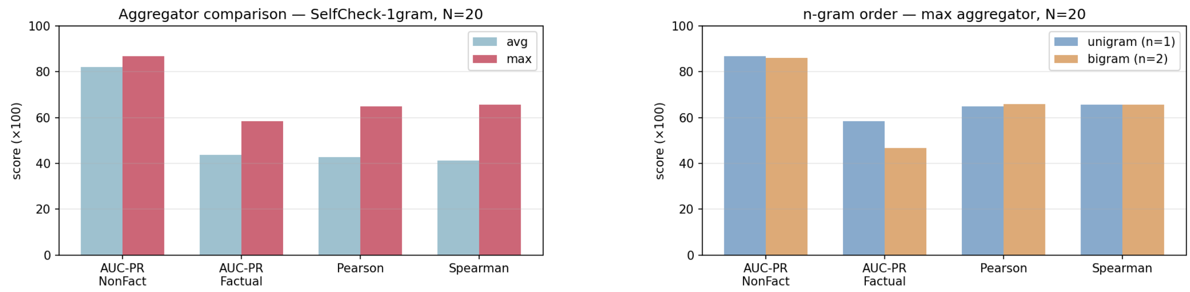

Figure 1: Ablations (SelfCheck-1gram, $N = 20$). Left: `max` vs. `avg` aggregator. Right: unigram vs. bigram order.

score near 7.0 (Fig. 2, right): GPT-2 hallucinates so heavily — even *Marie Curie* becomes a movie star — that no grounded baseline exists to contrast against. SelfCheckGPT assumes a competent generator, and silently fails otherwise.

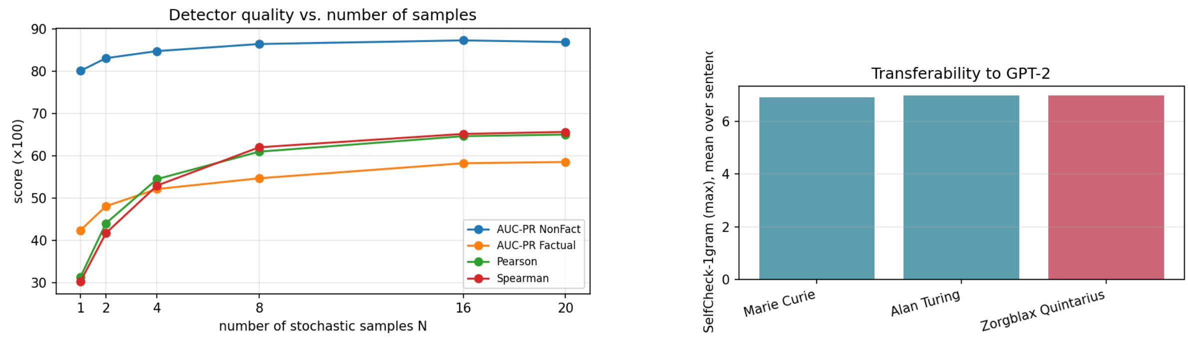

Figure 2: Left: detector quality vs. number of samples N (plateau near $N \approx 8$). Right: transferability to GPT-2 — all subjects, including the fabricated one (red), score near 7.0.

3.3 Comparison with the NLI method

Where the n-gram method detects inconsistency through **lexical overlap**, the NLI variant replaces this surface-level signal with **semantic entailment**. The intuition is the same — a factual sentence should be consistent across samples — but NLI can detect contradictions even when the wording differs entirely.

Concretely, for each sentence r_i and each sample S^n, we run a 3-class NLI classifier (`cross-encoder/nli-deberta-v3-large`, 435M parameters) with the sample as premise and the sentence as hypothesis. We renormalize over only the entailment and contradiction logits, discarding the neutral class (Eq. 9 of the paper), and average the contradiction probability across all $N = 20$ samples (Eq. 10 of the paper).

Our results on the full 238-passage benchmark are shown in Table 1. NLI outperforms n-gram on **sentence-level detection** (+2.8 AUC-PR on NonFact), confirming that semantic inference catches hallucinations that lexical overlap misses. However, n-gram edges ahead on **passage-level correlation** (Pearson: 65.0 vs. 63.1), suggesting that mean contradiction probabilities aggregate less smoothly than max n-gram surprises. Our NLI reproduction falls about 11 points below the paper's NLI numbers on the correlation metrics (Pearson 63.1 vs. 74.1), likely due to unspecified inference-time details; our n-gram, by contrast, closely matches the paper's n-gram numbers (Pearson 65.0 vs. 64.7).

Metric	NLI (ours)	Paper	1gram-max (ours)
AUC-PR NonFact	**89.74**	92.50	86.90
AUC-PR Factual	**59.37**	66.08	58.90
Pearson	63.09	74.14	**65.00**
Spearman	61.37	73.78	**65.90**

Table 1: Results on the full 238-passage benchmark ($N = 20$).

In summary, NLI offers stronger sentence-level detection at a higher computational cost, while n-gram remains competitive at the passage level with no GPU dependency. Our results confirm the paper's ranking, though the gap between the two methods is smaller than the paper suggests.

4 Limitations

4.1 Paper Limitations

Several limitations of SelfCheckGPT are worth highlighting, some of which the authors acknowledge themselves.

Evaluation Scope The entire evaluation is conducted on a single domain, biographical texts about individuals, and using a single model, GPT-3. This makes it hard to conclude whether the method generalizes well to other types of content (scientific texts, news articles, or open-ended reasoning), other generation styles, or more recent model families. A broader evaluation across domains and models would have significantly strengthened the paper's claims.

Consistency assumption can fail The core assumption of SelfCheckGPT is that hallucinated facts will vary across samples while true facts will remain consistent. However, this assumption does not always hold. If a false piece of information is well-represented in the model's training data, the model may hallucinate the same incorrect fact consistently across all samples, causing SelfCheckGPT to incorrectly classify it as factual. This is a fundamental limitation of the approach that is not discussed in the paper.

Sentence-Level granularity The method evaluates factuality at the sentence level, but a single sentence can contain both factual and non-factual information simultaneously. For instance, a sentence may correctly name a person but wrongly state their birth date. This limitation is briefly acknowledged by the authors, who point to later work on atomic-level factuality evaluation as a natural extension.

Computational Cost Several variants, particularly the Prompt variant, require a large number of LLM calls, one per sentence per sample, making the method expensive to run at scale. The authors estimate around \$200 in API costs for evaluating their dataset of 1908 sentences with 20 samples using GPT-3, and around \$20 with ChatGPT. While the NLI variant is more affordable, the overall sampling requirement (N calls per query) multiplies inference costs significantly compared to a single-pass approach.

No real-time applicability Because SelfCheckGPT requires generating N additional responses for every query, it cannot easily be used in latency-sensitive applications. This practical constraint is not discussed in the paper, yet it is an important consideration for real-world deployment.

4.2 Contribution Limitations

Beyond the inherent limitations of the original study discussed above, our own implementation and experimental setup also present certain shortcomings.

First, while the SelfCheckGPT framework proposes five different variants to measure consistency, we only implemented and evaluated two of them. By restricting our focus solely to the n-gram and NLI variants, we cannot independently confirm the reproducibility, exact performance, or computational costs of the BERTScore, QA, and Prompting methods.

Furthermore, regarding the NLI method we did implement, we observed an unexplained gap of approximately 10 points on the correlation metrics compared to the results reported by the authors. As noted in our results, we strongly suspect this discrepancy arises from unspecified inference-time details in the original paper, but this limitation remains unresolved in our current evaluation.

Finally, our original experiment designed to test the transferability of the method to a smaller model was heavily restricted in scope, as the GPT-2 evaluation only covered three subjects. Although this small sample was sufficient to demonstrate that the detection method fails silently when a model hallucinates entirely and consistently, a larger-scale evaluation with a more diverse set of subjects would be necessary to draw robust and definitive statistical conclusions.

5 Conclusion

In conclusion, SelfCheckGPT is a simple and effective method for detecting hallucinations in any black-box model without external resources. Our from-scratch implementation successfully validated the paper's core claims, confirming that the unigram variant is surprisingly competitive despite its simplicity. Furthermore, our comparative analysis established that the NLI variant offers the best cost-to-performance trade-off among the practical options.

Our experiments also yielded a valuable practical insight: using just N=8 stochastic samples is sufficient to maintain detection quality, effectively halving the computational cost with a negligible performance loss. However, a critical warning must be issued regarding the framework's limitations. As demonstrated by our transferability experiment, the method fails silently when the text generator is too weak or when specific hallucinations are deeply anchored in the training data. Ultimately, while SelfCheckGPT is a powerful tool, it strictly relies on the assumption of a highly capable and consistent generator.

6 Use of AI for this report

Throughout this project, we used artificial intelligence as a working assistant. In particular, it helped us clarify certain complex concepts and aspects of the research paper. During the implementation phase, its use remained highly controlled. We only relied on it for specific technical points, such as debugging.